Relational Atlas Framework

Exact Johnson–Type Transport Core

Naming Convention

This document uses **Relational Atlas Framework (RAF)** for the mathematical and combinatorial layer of the work.

RAF refers to:

- admissible comparison,
- discrete state spaces,
- atlas structures,
- Johnson–type geometries,
- shell observables,
- exact drift and diffusion laws,
- generating–function counting,
- folded quotients,
- and microscopic–to–coarse transport.

Emergent Horizon Framework (EHF) refers to the broader interpretive hypothesis that RAF–type structures may eventually help explain emergent geometry, gravitational consistency, horizon–like behavior, or other physical phenomena.

This document is primarily a RAF document. Its exact mathematical claims do not require EHF to be physically correct.

Core Principle

Only distinctions that admit admissible comparison are structurally meaningful.

Equivalently:

> When admissible comparison fails, distinctions collapse.

The Relational Atlas Framework studies how structured geometry, invariants, shell observables, and effective transport laws emerge from local admissible comparison on discrete state spaces.

This document presents the current mathematical core of the framework: an exact microscopic–to–coarse transport theory on Johnson–type subset geometries.

No physical interpretation is assumed.

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1. Emergence from Hamilton–Cycle Admissibility

The Johnson-type state space used in this document is not introduced as an arbitrary external structure.

It arises as the reduced comparison geometry of the Hamilton-cycle admissibility construction.

Starting from canonical Hamiltonian cycles with local circular reversal moves, strict admissibility produces parity-sector structure. In the even case, once the parity sector is isolated and an anchor/gauge convention is fixed, the relevant sector data can be encoded as a fixed-size subset

$$
S\subset [2k-1], \qquad |S|=k.
$$

Under this encoding, elementary admissible sector changes become single-swap moves

$$
S\longrightarrow S',
\qquad
|S\cap S'|=k-1.
$$

This is precisely Johnson adjacency.

Thus the Johnson geometry is not assumed as primitive. It is the effective comparison geometry induced by local Hamilton-cycle admissibility after sector reduction and subset encoding.

The full raw Hamilton-cycle move graph contains more information than the Johnson graph. The Johnson schema appears at the reduced/coarse comparison level.

The emergence chain is:

$$
\text{Hamilton-cycle local admissibility}
\rightarrow
\text{parity-sector reduction}
\rightarrow
\text{subset encoding}
\rightarrow
\text{Johnson geometry}.
$$

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1. Overview

The cleanest current realization of the framework is the even-side Johnson-type state space

$$
\Omega_k = \{S \subset [2k-1] : |S| = k\}.
$$

Adjacency is given by the single-swap Johnson move:

$$
S \longrightarrow S' \quad \text{where } |S \cap S'| = k-1.
$$

Equivalently, one element is removed from the subset and one new element is inserted.

The main shell observable is

$$
h(S)=\sum_{i\in S} i.
$$

A secondary observable is

$$
q(S)=\sum_{i\in S} i^2.
$$

The central result is that this system admits exact microscopic drift and diffusion laws, and that the shell-level transport law is an exact low-moment coarse-graining of the underlying subset-counting theory.

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2. State Space and Shell Structure

2.1 State space

For fixed positive integer k, define

$$
\Omega_k = \binom{[2k-1]}{k}.
$$

That is, each state is a k-element subset of the label set

$$
[2k-1]=\{1,2,\dots,2k-1\}.
$$

The number of states is

$$
|\Omega_k|=\binom{2k-1}{k}.
$$

2.2 Johnson adjacency

Two states $S,T\in\Omega_k$ are adjacent when they differ by one element:

$$
|S\cap T|=k-1.
$$

This is the standard Johnson graph adjacency on $J(2k-1,k)$.

Each state has degree

$$
d=k(k-1),
$$

because there are k choices of element to remove and $(2k-1)-k=k-1$ choices of element to insert.

2.3 Shell observable

The shell coordinate is

$$
h(S)=\sum_{i\in S} i.
$$

The central shell is

$$
h_\ast = k^2.
$$

The exact mean and variance over the uniform state space are

$$
\mathbb E[h]=k^2,
$$

and

$$
\operatorname{Var}(h)=\frac{k^2(k-1)}{6}.
$$

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3. Exact Drift Law

Define the one-step shell increment by

$$
\Delta h = h(S')-h(S),
$$

where S' is chosen uniformly among the Johnson neighbors of S.

The shell drift is

$$
A(h)=\mathbb E[\Delta h\mid h(S)=h].
$$

The exact drift law is

$$
A(h)=\frac{2k-1}{k(k-1)}(k^2-h).
$$

Consequences:

- the drift is exactly linear in shell height,
- the restoring center is exactly $h_\ast=k^2$,
- states below the center drift upward on average,
- states above the center drift downward on average,
- the shell dynamics has an exact central restoring tendency.

This is not a numerical fit. It is an exact combinatorial transport law.

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4. Exact State-Level Diffusion Law

Define the second shell increment moment by

$$
B(S)=\mathbb E[(\Delta h)^2\mid S].
$$

Introduce

$$
T_1=\sum_{i=1}^{2k-1} i = k(2k-1),
$$

and

$$
T_2=\sum_{i=1}^{2k-1} i^2 = \frac{(2k-1)(2k)(4k-1)}{6}.
$$

For

$$
h(S)=\sum_{i\in S}i,
$$

and

$$
q(S)=\sum_{i\in S}i^2,
$$

state-level diffusion is exactly

$$
B(S)=\frac{kT_2-q(S)-2T_1h(S)+2h(S)^2}{k(k-1)}.
$$

Thus diffusion depends exactly on the pair (h,q).

Shell height alone does not determine the microscopic diffusion. The first exact correction beyond shell height is carried by the second moment observable q.

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5. Exact Shell Diffusion

The shell diffusion is obtained by averaging the state-level diffusion over all states in a shell:

$$
B(h)=\mathbb E[B(S)\mid h(S)=h].
$$

Using the state-level law, this gives

$$
B(h)=\frac{kT_2-\mathbb E[q\mid h]-2T_1h+2h^2}{k(k-1)}.
$$

Therefore the shell-level diffusion is exact once the conditional second moment

$$
\mathbb E[q\mid h]
$$

is known.

This establishes the shell transport law as a genuine low-moment coarse-graining of exact microscopic dynamics.

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6. Generating-Function Counting Layer

The shell and moment laws are supported by an exact subset-counting theory.

Define

$$
G_k(x,y)=\prod_{m=1}^{2k-1}(1+yx^m).
$$

The shell count is

$$
N_h=[x^h y^k]G_k(x,y).
$$

Thus N_h counts the number of states $S\in\Omega_k$ with shell height $h(S)=h$.

For label-refined counts, define

$$
F_{k,j}(x,y)=\prod_{m\ne j}(1+yx^m).
$$

Then

$$
N_{h,j}=[x^{h-j}y^{k-1}]F_{k,j}(x,y).
$$

Here $N_{h,j}$ *counts the number of states in shell* h *that contain label* j.

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7. Exact Shell Moment Identities

The label-refined counts satisfy the exact identities

$$
\sum_j N_{h,j}=kN_h,
$$

$$
\sum_j jN_{h,j}=hN_h,
$$

and

$$
\sum_j j^2N_{h,j}=N_h\,\mathbb E[q\mid h].
$$

These identities show that shell-conditioned first and second moments are exact coefficient-level consequences of the generating function.

The hierarchy is therefore:

1. exact subset-count generating function,

2. exact shell counts,
3. exact label-refined counts,
4. exact shell moment identities,
5. exact microscopic drift and diffusion,
6. exact shell coarse transport.

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8. Exact Reflection Law

The label-refined shell counts satisfy

$$
N_{h,j}=N_{2k^2-h,\,2k-j}.
$$

This reflection law induces mirror symmetry around the central shell

$$
h_\ast=k^2.
$$

It explains the observed reflection structure in:

- shell counts,
- label occupancies,
- shell-conditioned second moments,
- shell diffusion profiles.

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9. Effective Fokker–Planck Interpretation

At shell level, the system admits an effective one-dimensional transport description using the exact drift $A(h)$ and exact shell diffusion $B(h)$.

This may be written in Fokker–Planck language, but the interpretation should be precise:

- it is not a fundamental continuum theory,
- it is an exact coarse-grained shell transport law,
- it becomes smoother and more continuum-like as k grows,
- its coefficients are derived from exact finite combinatorics.

The shell-level picture is therefore an effective transport bridge from finite subset dynamics to continuum-style drift-diffusion language.

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10. Exact Versus Effective Results

10.1 Exact results

The following are exact on the even Johnson-type state space:

- the state space $\Omega_k=\binom{[2k-1]}{k}$,
- Johnson single-swap adjacency,
- the shell observable $h(S)$,
- the second moment observable $q(S)$,
- the central shell $h_\ast=k^2$,
- the shell mean $\mathbb E[h]=k^2$,
- the shell variance $\operatorname{Var}(h)=k^2(k-1)/6$,
- the drift law
  $$A(h)=\frac{2k-1}{k(k-1)}(k^2-h),$$
- the state diffusion law
  $$B(S)=\frac{kT_2-q(S)-2T_1h(S)+2h(S)^2}{k(k-1)},$$
- shell diffusion via $\mathbb E[q\mid h]$,
- the shell count generating function,
- the label-refined generating function,
- the shell moment identities,
- the reflection law
  $$N_{h,j}=N_{2k^2-h,\,2k-j}.$$

10.2 Effective or unresolved results

The following remain effective, approximate, or open:

- simple closed formulas for $\mathbb E[q\mid h]$,
- low-degree closures for full label profiles $N_{h,j}$,
- recurrence relations for the full label-refined count arrays,
- asymptotic continuum limits beyond the exact shell law,
- broader classification of state spaces with comparable moment closure.

A quadratic approximation for $\mathbb E[q\mid h]$ appears highly accurate and improves with k, but it is not yet the exact full count law.

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11. Full-Distribution Complexity and Low-Moment Simplicity

The full shell-conditioned label distribution $N_{h,j}$ is structurally rich. It does not appear to collapse to a simple low-degree polynomial law in (h,j).

However, the low moments are clean and exact.

This creates an important distinction:

- the microscopic distribution remains complex,
- the low-moment transport law is simple,
- the shell-level effective dynamics is therefore a true coarse-graining, not a trivial rewriting of the full distribution.

This is a central structural feature of the framework.

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12. Folded Odd-Side Extension

A natural folded counterpart begins from

$$
\widetilde{\mathcal X}_{\mathrm{odd}}(k)=\binom{[2k]}{k},
$$

with complement identification

$$
S\sim S^c.
$$

The folded state space is

$$
\mathcal X_{\mathrm{odd}}(k)=\widetilde{\mathcal X}_{\mathrm{odd}}(k)/(S\sim S^c).
$$

This side appears structurally nontrivial but is less complete than the even theory.

Current status:

- quotient geometry is natural,
- folded shell behavior has been observed,
- representative-flip effects appear in longer loops,
- simple parity predictors are weak,
- a simple $\mathbb Z_2$ edge-cocycle explanation does not yet cleanly close the theory.

At present, the strongest exact results belong to the even Johnson-type transport system.

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13. Current Mathematical Status

The framework currently supports the following precise statement:

> There exists a Johnson-type relational state space with local admissible comparison whose microscopic transport laws are exact, whose shell-level drift and diffusion are exact low-moment projections, and whose coarse dynamics is rooted in exact subset-count generating functions.

This establishes a finite combinatorial route from local moves to effective transport laws.

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14. Working Conjecture

Exact Low-Moment Realization Conjecture

There exists a broader class of relational state spaces with local admissible comparison such that:

1. exact microscopic transport observables can be defined,
2. exact low-moment shell dynamics emerges from them,
3. shell-level effective transport can be derived as a true coarse-graining,
4. the full microstate distribution remains richer than the low-moment closure,
5. different realizations may differ mainly by geometry, quotienting, or admissibility rules.

The even Johnson-type subset geometry is the first clean exact realization of this program.

The folded odd-side constructions are candidate further realizations, but remain incomplete.

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15. Immediate Research Directions

Immediate directions:

- derive closed formulas for $\mathbb E[q\mid h]$,
- derive higher shell moments from generating functions,
- determine which moments close cleanly and which do not,
- search for asymptotic laws for shell-conditioned moment hierarchies,
- extend or obstruct the folded odd-side theory,
- classify other admissible-comparison spaces with exact low-moment transport.

Longer-term directions:

- formulate the theory abstractly beyond Johnson graphs,
- identify universality classes of admissible-comparison transport,
- determine whether folded quotients form a distinct transport class,
- investigate links to statistical mechanics, emergent geometry, and information-theoretic physics without assuming such interpretations in advance.

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16. Summary

The Relational Atlas Framework currently has a clean mathematical core:

- a Johnson-type subset state space,
- local single-swap admissible comparison,
- exact shell observables,
- exact microscopic drift,
- exact state-level diffusion,
- exact shell diffusion via conditional second moments,
- exact generating-function shell counts,
- exact label-refined counts,
- exact reflection symmetry,
- and an exact microscopic-to-coarse transport bridge.

The even Johnson-type system is currently the benchmark realization.

The folded odd-side system is the next major extension.

The framework should presently be understood as exact combinatorial transport theory. Any physical interpretation remains a separate later step.

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17. Relationship to the Emergent Horizon Framework

The Emergent Horizon Framework is best understood as a broader interpretive layer built around the possibility that admissible-comparison structures may underlie emergent physical geometry.

The Relational Atlas Framework supplies the mathematical core for that possibility.

In this relationship:

- RAF provides exact combinatorial structures,
- RAF defines admissible comparison and local transport,
- RAF derives shell observables and coarse laws,
- RAF studies atlas and quotient structures without assuming physics,
- EHF asks whether such structures can be interpreted physically.

The exact Johnson-type transport results strengthen the mathematical foundation available to EHF, because they show that local admissible comparison can produce:

- nontrivial state-space geometry,
- exact microscopic transport laws,
- exact coarse shell dynamics,

- reflection structure,
- and an emergent drift–diffusion description.

However, these results do not by themselves establish a physical theory.

They do not yet identify:

- what the states physically represent,
- what the shell coordinate corresponds to in nature,
- what physical observable is represented by h,
- what physical observable is represented by q,
- or which empirical system, if any, realizes this structure.

The correct current status is therefore:

> RAF is the exact mathematical framework.
> EHF is the broader physical hypothesis that RAF-type structures may help explain emergent geometry or horizon-like physics.

The Johnson transport core should be presented first as RAF. EHF should be introduced only after the exact mathematical structure has been clearly stated.